

Dejan Grubisic

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EXPERIENCE

Axiom Math

Founding Engineer

Palo Alto, CA

Jun. 2025 – Jul. 2026

- Led an engineering team to deliver a large-scale autoformalization pipeline
- Engineered an AI system for code optimization, formally verified in Lean
- Built Axiom's first agentic system for automated theorem proving in Lean
- Developed a system for scalable data collection and contributed to model training

Meta - Modern Recommendation Systems

AI Research Scientist

Menlo Park, CA

Jul. 2024 – Jun. 2025

- Worked on Facebook Feed foundation models for ranking and content understanding
- Profiled GPU utilization/memory to guide feed ranking training efficiency

Meta - Fundamental AI Research (FAIR)

Machine Learning Research Intern

Menlo Park, CA

May 2023 – Dec. 2023

- Trained a 7B Llama 2 for LLVM compiler optimization, reducing code size 3.0% beyond -Oz
- Introduced Priority Sampling, raising model gains over -Oz from 2.87% to 5% and beating the autotuner
- Ablated 3 compiler-feedback mechanisms and showed that simple sampling performs better

Meta - Fundamental AI Research (FAIR)

Machine Learning Research Intern

New York, NY

May 2022 – Dec. 2022

- Built LoopTune, an RL tensor-optimization framework: 9.7x faster than TVM, 2.8x MetaSchedule
- Trained an RL policy reaching within 3% of hand-tuned Numpy, while tuning in ~1s vs 33-62s for autotuners

Rice University

Graduate Research Assistant

Houston, TX

Aug. 2019 – May 2024

- Developed GPU support for HPCToolkit, a large-scale performance profiler
- Created performance analysis techniques for GPU-accelerated applications
- Developed profiler-guided optimization based on reinforcement learning (Compiler2)

Berkeley Lab

High Performance Computing Research Intern

Berkeley, CA

Jun. 2021 – Sep. 2021

- Profiled and analyzed power consumption of multi-node GPU applications

EDUCATION

Doctor of Philosophy | High Performance Computing and Machine Learning

Rice University

Aug. 2019 – May 2024

Houston, TX

Master of Science | Big Data Architectures

University of Novi Sad | GPA: 10.00 / 10

Aug. 2018 – May 2019

Novi Sad, Serbia

Bachelor of Science | Electrical and Computer Engineering

University of Novi Sad | GPA: 9.98 / 10

Aug. 2014 – May 2018

Novi Sad, Serbia

SELECTED PUBLICATIONS

- D. Grubisic et al. Compiler Generated Feedback for Large Language Models. *arXiv preprint*, 2024.
- D. Grubisic et al. Priority Sampling of Large Language Models for Compilers. *EuroMLSys @ EuroSys*, 2024.
- D. Grubisic et al. LoopTune: Optimizing Tensor Computations with Reinforcement Learning. *arXiv preprint*, 2023.
- C. Cummins, V. Seeker, D. Grubisic, et al. Large Language Models for Compiler Optimization. *arXiv preprint*, 2023.
- B. Wasti, D. Grubisic, et al. LoopStack: ML-friendly ML Compiler Stack. *ML for Systems Workshop @ NeurIPS*, 2022.
- K. Zhou et al (incl D. Grubisic) An Automated Tool for Analysis and Tuning of GPU-Accelerated Code in HPC Application. *TPDS*, 2022.
- K. Zhou et al (incl D. Grubisic) Measurement and Analysis of GPU-Accelerated Applications with HPCToolkit. *Parallel Computing*, 2021.

SELECTED PROJECTS

- Master Thesis** - Multi-source shortest paths on dynamic graphs via Lambda architecture (pySpark, HDFS, Kafka, Docker)
- Bachelor Thesis** - FPGA core accelerating a chess engine; ESL design in SystemC, VHDL board evaluation, UVM verification
- Parallel Computing** - Game-tree search (Cilk), LU decomposition (OpenMP), 2.5D matmul (OpenMPI), bitonic sort (CUDA)

COMMUNITY AND LEADERSHIP

Southeast Europe AI Summit - Founder, connecting AI ecosystem representatives from 7 countries across Southeast Europe

May. 2026

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA C, Lean, Java, Bash, VHDL

Systems: PyTorch, LLVM, OpenMP, MPI, CUDA, Docker, Slurm, Spark, Linux

AI/ML: LLMs, fine-tuning, reinforcement learning, data preparation, compiler construction, parallel computing, formal verification